

Micah Shiels

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Location: Spain | Right to Work: EU & UK (Unlimited)

SUMMARY

Backend Software Engineer with almost four years' experience designing and optimising AI-powered backend systems in Python. Strong background in FastAPI, PostgreSQL, Google Cloud Vertex AI and performance engineering, with experience delivering scalable production services and improving database and data processing performance.

EXPERIENCE

Professional Development | Upskilling | February 2026 – Present | Remote

- Built and deployed a personal portfolio website (micah.es).
- Completed advanced training in System Design, Software Architecture, SOLID principles and Data Structures & Algorithms.
- Continued developing German and Portuguese language skills.

Mid Software Engineer | StudyHall AI | July 2022 – February 2026 | Remote (UK-based company)

- Re-engineered a production Vertex AI-based EPUB processing pipeline by introducing Redis caching, eliminating race conditions, optimising SQL queries and database connections, migrating data processing from Pandas to Polars, and removing redundant data loading, **significantly reducing processing time, increasing throughput, and enabling more EPUBs to be processed concurrently.**
- Independently maintained and optimised the company's primary PostgreSQL database, reducing execution time of complex SQL queries by approximately **90%**.
- Developed and optimised FastAPI REST backend services supporting planning, assessment, reporting, module generation and examination workflows **used by thousands of users.**
- Designed and implemented Python data ingestion pipelines replacing legacy VBA workflows, reducing content ingestion time by approximately **90%**.
- Optimised and maintained a TypeScript Postgres to MongoDB seeder used by the client team heavily **reducing runtime of full seeder run from hours to minutes**, mainly by identifying and refactoring N+1 queries.
- Developed internal tools using Retool and Streamlit, automating repetitive workflows for the Curriculum / Data team and aided in decision-making through dashboards.
- Mentored junior engineers through code reviews and technical guidance, promoting engineering best practices and improving code quality.

English Teacher (TEFL) | Your Language Club Getafe | October 2021 – July 2022 | Madrid, Spain

- Delivered over 400 English lessons, strengthening communication skills and the ability to explain complex concepts to diverse audiences.

Database Administrator | Ministry of Furniture | October 2017 – October 2021 | Neath, UK

- Independently maintained and optimised MySQL and Microsoft SQL Server databases supporting over 6,000 products and 300,000+ product variants.
- Reduced annual catalogue update time by approximately 80% by automating manual processes using SQL and VBA.

SKILLS

- **Computer Languages:** Python, SQL, PL/SQL, TypeScript, JavaScript, VBA, HTML, CSS, Neo4j Cypher
- **Frameworks:** FastAPI, Pydantic, Numpy, Pandas, SQLAlchemy, SQLAlchemyModel, BeautifulSoup, Django, Astro, Tailwind
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Neo4j
- **Tools:** GitHub, Docker, Google Cloud, Sentry, Retool, Streamlit, JIRA, Claude Code, Gemini CLI, Ollama, Codex, Cursor
- **Natural Languages:** English (Native), Spanish (C2), Portuguese (B1), German (A1)
- **Certifications:** DELE C2 Spanish, TEFL Distinction